

ibm3081_diagnostic_sop_v1json

```
{
  "id": "SOP-IBM3081-001",
  "title": "IBM 3081 Processor Unit: System Check & Error Logging",
  "system": "IBM 3081",
  "component": "3081 Central Processing Unit (CPU) / Service Processor",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Maintenance Library (SY22-2841 / SY33-1057)",
  "description": "Diagnostic procedure for analyzing system checks and service processor error logs for the IBM 3081 Processor Complex.",
  "steps": [
    {
      "step": 1,
      "action": "Service Processor (SVP) Status Review",
      "details": "Access the Service Processor console. Check for any 'Hard Machine Check' (HMC) or 'Soft Machine Check' (SMC) entries in the error log.",
      "indicator": "HMC indicates a catastrophic failure requiring immediate hardware replacement."
    },
    {
      "step": 2,
      "action": "Instruction Address Stop (IAS) Analysis",
      "details": "Display the Instruction Address Register (IAR) on the SVP console. Record the address where the processor stopped.",
      "check": "Cross-reference the IAR with the microprogram map to identify the failing instruction sequence."
    },
    {
      "step": 3,
      "action": "Check Bit Verification",
      "details": "Examine the status of the 'Machine Check' and 'Channel Check' bits in the System Status Word.",
      "logic": "If Channel Check is active, investigate the I/O interface drivers and the associated channel logic board."
    },
    {
      "step": 4,
      "action": "Fault Location Test (FLT) Loading",
      "details": "Initiate FLT loading from the system diskette drive. These tests are controlled by the Service Processor.",
      "remedy": "The FLT result code will identify the specific board or thermal conduction module (TCM) that is failing."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "SVP fails to boot",
      "next": "Check SVP power supply and console cable connectivity."
    },
    {
      "condition": "Instruction Damage bit ON",
      "next": "Run CPU Logic FLTs to isolate failing TCM."
    }
  ]
}
```

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    {
      "condition": "Error is intermittent",
      "next": "Run 'Loop on Error' micro-diagnostics to verify logic stability."
    }
  ]
}
```